Division of Information Technology & Sciences
Department of Computer & Digital Forensics
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LongForm Report

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~~~~~~ Table Of Contents ~~~~~~~

Project Description:

This project aims to implement a self created IAC [Infrastructure as Code] automation platform, for proxmox clustered hosts, to deploy enterprise level environments with inbuilt and automatable malware remediation of multiple known samples. This project combines hardware, infrastructure engineering, malware analysis, and automation for streamlining enterprise security operations. Automated scripts use Ansible and powershell specifically. The samples presented in this project consist of, but are not limited to, WannaCry, Loki Bot, Agent Tesla, et;all. This will be done within a localized environment hosted by Connor East. Hardware specifications will be specified in the “Hardware Section”. The end goal of this project is to have a fully deployable IAC VM for deployment on the cyber.local and df.local environments respectively should it be deemed to be useful within the team's project scope.

Required Dependencies:

Below are two sections, the first will relate to the hardware required for project functionality. The contents of this section will not include dependencies required on the Virtual Machines. This only includes content required for access.

Hardware:

Below is a list of the devices being hosted by Connor East for the purpose of this project. Items, unless otherwise stated to have been borrowed, were bought by Connor East for the purposes required by this project.

Hardware	**Purpose.**
TP-Link AX5400 Wifi 6 Router	Given the environment is not being hosted by Champlain, a router was required to be used as a default gateway. The prior gateway which had been placed on the premises was an ADTRAN 854-v6 ISP. The ADTRAN 854-v6 worked as a router but did not allow for the modularity required for this project. Specific errors being failures with the web terminals ability to interface with the router and open connections for cloudflare and wireguard tunneling
Pre-owned Catalyst 2960-CX Distribution Switch	Originally given to the team for the purpose of segmenting network traffic, this switch allowed each server to be ethernet joined. This improved on and off site access speeds significantly. The original software on this switch needed to be wiped and reinstalled. For more information on this see our Hardware setup log.
Kasa Smart Wifi Plug	This was bought specifically for the sole purpose of reducing electrical consumption. By turning on the servers only when needed; electrical consumption, and the attack surface were minimized.
Refurbished Dell PowerEdge R430	This 13th generation Dell server is to be used as the main cluster Proxmox server. Its main purpose will be for holding major infrastructure which needs to be maintained.

	This includes any tunnels, VPNs, Scripts, et;all. All major hardware is hosted here as well as all of our sysprepped VM templates. [See Research methodology section for more information]
Custom Server	This device is used for the sole purpose of minimizing quorum errors, Should a server, on a two server node, be unable to start quorum would fail and a consensus on what should and what should not be turned on won't be reached. This server has the following hardware: --- MotherBoard Manufacturer: Gigabyte Technology Co., Ltd. --- MotherBoard Model: B365M DS3H --- MotherBoard chipset: Intel B365
Cisco UCS C250 M1	This device, created in 2009 and donated to our team by Samuel Guinther, is our main testing environment. It is network segmented and does not contain any ability to communicate with any network at a non proxmox-host level. This server reached its end of life in 2011 and as such is considered outdated. Given its ability to handle heavy workloads this was assigned as our testing environment.
USB-to-SATA Adapter	The device “Cisco UCS C250 M1” had a dead cmos battery. Instead of replacing this to allow for the installation of proxmox; it was decided to remove the contents of the old drive instead. This being a useful skill and tool to have for data extraction over the long term.
4TB Sata Drive	The installation of a 4TB drive was required for this project as none of the devices had enough storage for mass scale enterprise level deployments.

SoftWare:

Below is a list of software and tools required for hardware setup and accessibility. Further details will be provided within the Lab-Setup-and-Equipment page.

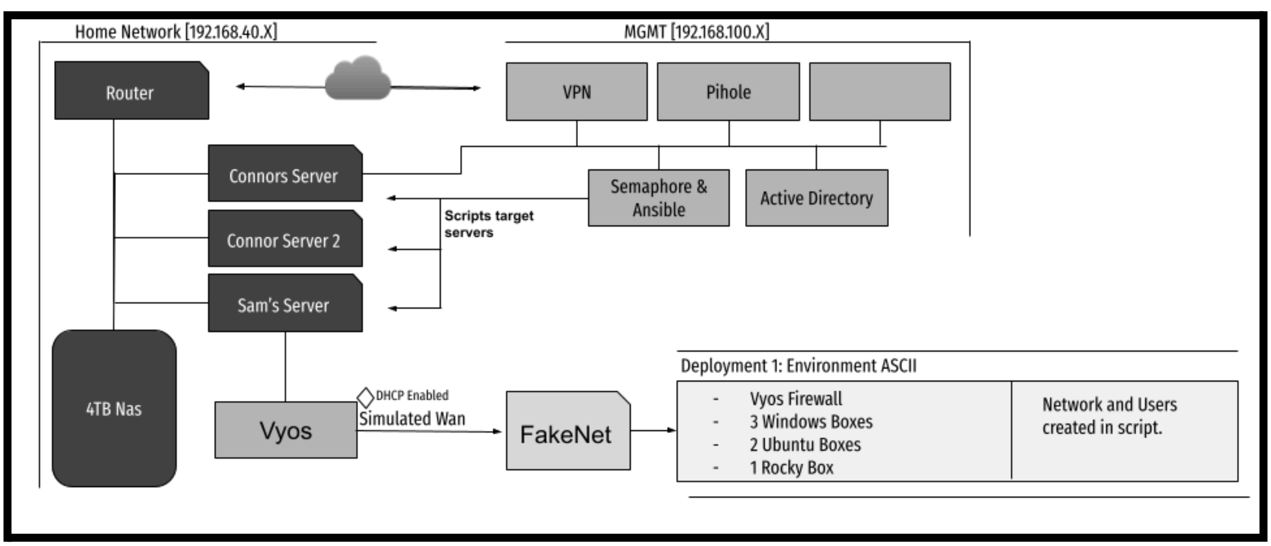
Software	**Purpose.**
CloudFlare Tunnel	A cloudflare tunnel and account were procured for remote access to the “Refurbished Dell PowerEdge R430” VPN box. This was required for Seraphim and Andrei to be able to access the content from campus once applicable.
Domain: [ceimprovment.work]	In order to set up the cloudflare tunnel a domain needed to be registered as such Connor registered

	ceimprovment.work for a three year lease. Connection to the proxmox based environment being accessible from https://serverconnection.ceimprovments.work/ when the server is online.
Cisco switch software	In order to implement the Catalyst 2960-CX Distribution Switch into our environment; a new software installation was required. Installation being completed using Rommon
4TB Nas Connection	Within the proxmox interface the 4TB drive, installed by Connor East, was set up as a NFS share. It is accessible by all host devices and is hosting both the base VM templates as well as the deployed environments.

Infrastructure Documentation:

Infrastructure is set up into three sectors; MGMT, Home, & Deployment. Home being Connor East's local home network where the infrastructure is located. MGMT being all of the VM's hosted on Connors Server [Dellr430]. Deployment being the VM's, deployed using our local ansible scripts, on Sam's Server. As of now we have a total of 14 sysprepped virtual machine templates, each of which conforming to

Figure 1: Network Map



The Network IP is visible for each section outside of ASCII. The purpose of this being that some of our deployment related scripts will actively assign IP's through the use of the proxmox qemu-agent; This has been found to work approximately 70% of the time. For the purposes of this capstone however, we will be using DHCP allocation for our Malware remediation environment. The FakeNet network being located on 10.10.10.X and the deployment environment being held on 192.168.10.X. For more thorough networking information please refer to the documentation labeled “BaseVM Tracking”. Should you wish to learn more about the actual setup of the server and all of its dependencies please view “Server Setup Documentation”

Literature Review:

Prior to beginning our project we were required to conduct a literature review. Over the span of three days information was gathered from several sources on tools similar to what we planned on creating, VM compliance benchmarks, as well as several instances of malware related academic literature.

Tools:

For tools we reviewed Terraform, Pulumi, Ansible, Puppet, Vagrant, Veewee, Red Hat Satellite; as well as the tools Zabbix, Nagios, and Manage Engine. Each of these tools led the way in how we have developed our suite of scripts. The use of our pvesh commands borrows the patterns for dynamic vm discovery much in the same way as the Terraform platform. The use of embedded python blocks within our scripts mimicking Pulumni’s philosophy of using full languages within calls.

From Puppet and Chef we have been working to achieve the same goal of compliance and malware remediation. The major difference being that instead of requiring a persistent agent service for moderation we use the qemu-agent, which is required for a lot of proxmox functionality, to bypass that requirement. Pulling from Satelite’s playbook on remote system remediation we built our scripts with that in mind. Our scripts even work if the VM is disconnected from the network; given the qemu-agent has not been removed or corrupted.

Benchmarks:

The benchmark frameworks, ISO/IEC 27035, Mitre Att&ck, NIST CSF 2.0, and CIS were researched in tandem with the tools. They provided the foundation to our remediation and environmental creation. Starting with ISO/IEC 27035’s five-phase incident management lifecycle (Plan, Identify, Assess, Respond, Learn). Scripts relating to SIEM installation meeting the Identify and Assess criterium; scripts accessible through Semaphore being used for the Respond category, while each script creates documentation for us to learn from in the form of logging. MITRE att&ck was used specifically in order to map artifacts left over on the system to their Malware family ID structure for easy comprehension. NIST CSF functions much in the same way as the ISO/IEC 27035 framework; however was implemented majoritively through our scripts for IP collection and Integrity Validation scripts. CIS was integrated specifically into our VM’s when they were set up; future projects may include automating the process through Semaphore.

Academic Literature:

Four separate pieces of academic literature were read, papers such as “Unleashing Full Potential of Ansible Framework: University Labs Administration” showing the practical application of ansible for Development Operations for the sole purpose of Idempotency. Idempotency being an IaC principle whose sole purpose is to reconfigure an environment to basal metrics as to achieve consistent configuration for environmental recreation, script deployments, et;all. While the paper “The practice of developing the academic cloud using the Proxmox VE platform” shows the platform's suitability for the case in which we were researching; a resource constrained environment which required reliable API-programmable virtualization with segmented architecture for security. This maps to our project through isolation on our malware analysis segment as well as our accessibility with tools such as cloudflare. The paper “An emerging threat Fileless malware: a survey and research challenges” described rule based detection, system behavioral detection, and several malware types which were useful of note for the implementation of our attempt at remediating Fileless Malware. The final paper we looked into was “Systematic Analysis of Infrastructure as Code Technologies“ which discussed the scripting languages most often used for IAC on proxmox as well as the ways in which they are often used.

Production Scripts:

Each script has been broken down into their corresponding classification. Deployment scripts are scripts used for creating a virtual environment; Integrity scripts are used to check the validity of the boxes; Remediation scripts are used to remove malware from a host; Siem scripts install System Integrity and Event Management tools on a specified box.

Deployment:

Script Name	**Purpose.**
deployment1-benchmark	This is a benchmark script focusing on DHCP based environmental creation. There are a total of five files being used in tandem for the deployment. User data is kept within the roster file. Configurations at the proxmox level are commented out specifically to allow for DHCP allocation
deployment1-remastered	This is a deployment script focusing on deploying a user environment and hardcoding the environmental structure using the qemu agent. This includes setting vyos, windows, and linux network configurations for a network generated in the setup_users.yaml file.
proxmox_eth5_assign.yaml	A script which assigns an IP based off of the attached bridge. This was used for testing purposes however it shows proficiency in scripting logic. Implementation of this occurred within deployment-remastered

VMRemovalByPool.yaml	This script removes VMS based on the group in which it is assigned. In this case it would be based on both the username and the network adapter they are connected to. These flags are set in the users Variable.

Integrity:

Script Name	Purpose.
ASCII-VMs-Sha256Collection.yaml	This collects all VM's with the appended variable "ASCII" in its name. This then takes the Sha256 hash of the qcow2 files before creating a json output in the vm-hashes directory. Future versions of this script should allow the user to choose based on group.
baseVMs-Sha256Collection.yaml	This collects all VM's within the allotted vmid range [3000-3100]. This then takes the Sha256 hash of the qcow2 files before creating a json output in the vm-hashes directory. Future versions of this script should allow the user to choose based on group.

Reconnaissance:

Script Name	Purpose.
IP_Collect.yaml	This script collects the IP's of all devices with an active qemu-agent installed. It runs through all the running vms before collecting their interface information. Following completion a consolidated report is templated and printed to the /tmp/ directory of awx
QemuAgent_Online.yaml	This script confirms the installation of, and addition of, the qemu-agent on all vms at the proxmox level.

Remediation:

Script Name	Purpose.
wannacry-remediation.yaml	This script remediates the malware "INSERT_HASH" which is a sample of the well known WannaCry Virus. This script removes all artifacts and processes created by this strand of malware. More information on the artifacts can be found in the malware remediation section of this paper. Only major note: the script does not decrypt wannacry files. This simply removes the malware from the system for fixing later.
conflicker-remediation.yaml	
agenttesla-remediation.yaml	

Siem:

Script Name	Purpose.
server.yaml	This installs velociraptor on a specified host device. You can choose which host you would like the server downloaded on.
agents.yaml	This installs the velociraptor client on all hosts on a specified network adapter. You can choose the network adapter in which you want to install velociraptor on. Minor issues exist within the installation stage, on windows, where the service can't start without a restart.

Deployment Hashes [Sha256]:

Below are the Sha256 hashes compiled using the “ASCII-VMs-Sha256Collection.yaml” and “baseVMs-Sha256Collection.yaml” integrity scripts created for this project.

VM Template Hashes

VMID	Name	SHA-256 Config Hash
3000	FakeNet-Base	61842a71aa16f5f2c814b8d263b390ad0ed2eb18276e32d9d19b1dd1a5e01005
3001	Windows10-Base	f750d697cda3203750e065835025463262a0340b513991f04db908acb6533dda
3002	Windows11-Base	d65f7f132520265b466bab5f11c4413b78c3a851feddda192224c4108b519881
3003	WindowsSRV-Base	ceb2c583e3b9a85be2e25ef5b7b3461fb40cadedbe88a701a9719df6a0bc1f2e
3004	UbuntuSRV-Base	fb191102b15f0d390ff3c692bf6e9087ed9be00e1dede956d0c56293621214f3
3005	UbuntuVM-Base	23c86986f706953fc7ec11f0feaa3cd33f4b83f7c73e2d0f1a6e2a94f341464c
3006	RockyVM-Base	17b8b829fe3e4853a1653b4d6a570c443cc2a408d427ae800c1d47424d3150a8
3007	KaliVM-Base	cac040f7183d43083514f72190eaa7c6dd4ba49b7561896df810346b228f9e91
3008	Vyos-Base	d071f811e9bae18a5968173df2c4b49481e9bfb8d1204f07104a92aca6d92da7
3009	PFSense-Base	d40f80a22f049c31745d5bee26f34f0b41b1c753d838a0f5e7e502ce06e0f674
3010	Vyos-GW	e18cc42ebc9e407c497efecf1949de474bd7a5c786dd7d942aef5e55504653fd
3011	Vyos-DHCP	fb2e970e8634479c70374cf90a77f79390d9978e3dfddf393e2887d3552d0180
3012	zeek-template	8d23acf9a115e2e5bb104ae87dbdbe750c4e6b2b1553a31d5e10fa1dd51e10a9

VMID	Name	SHA-256 Config Hash
3013	MalwareHost-Base	37e772e2377940e36846822f71fafb10621ea1fff2e7544bc918c4531b2d371f

Deployed VM Hashes

VMID	Template Name	SHA-256 Config Hash
2003	VyOS-ASCII	a106bf31bff0a0fcf2c15e557b7b5f859259a52cdfe439eba7c8a8e380a926a9
2004	Win10-1-ASCII	4ee22ea8c559fa87d093c4512bb6ed29d6300f89b2bb03c67d78ccb86bc755a9
2005	Win10-2-ASCII	4e87f28000f01fd8c4fe4ace9ea3b94c0cc20dc67136c77db0c90416528eaefe
2006	ubusrv-1-ASCII	520f7909be62bc094ba08a2d751de764a1f4f17d6b5ba9c65176187cd8aa794f
2007	ubusrv-2-ASCII	6143bf1cf8324bb734268cdff72ee136e418b7921f71fd8c47d2849ed7d3defd
2008	ubusrv-3-ASCII	e8a6b6302fc107609cd312ea887823fa44743fa8471ab9966328482e03794df6
2009	rocky-1-ASCII	4666b613e3bbc4d610b1f62e313a0f52241b72aef843b74182cbbf09b9de00b8
2010	rocky-2-ASCII	540ae97dd96dae5e20338827d278e6aa3274ac80edb067ed61c4a7d3d6804621
2011	kali-1-ASCII	d567f5ca64c9225079a245d66df0df1b0eba4ced6f6a6dee11bab8d262e3e08c
2012	kali-2-ASCII	a82a7f680749cb54cbe1c12b433983c590b1aa0a048c06da1b9a8967f2eeb468

Deployment Benchmarking:

Benchmarking was the main sector, outside of main template creation, of which Seraphim had been assigned. Both manual and Scripted deployments were measured. Below are our findings over five iterations of manual and automated deployments:

Scenario	*Method*	*Run*	*Total Time*	*Peak CPU (%)*	*Peak RAM (%)*
10 VM	Manual	1	2:59	1.12%	3.27%
10 VM	Manual	2	2:34	1.67%	3.28%
10 VM	Manual	3	2:54	1.68%	3.30%
10 VM	Manual	4	2:02	1.08%	3.25%
10 VM	Manual	5	1:40	1.32%	3.28%
10 VM	Script	6	1:12	0.38%	2.99%
10 VM	Script	7	1:12	1.33%	2.98%

10 VM	Script	8	1:13	0.42%	3.0%
10 VM	Script	9	1:12	1.21%	2.99%
10 VM	Script	10	1:12	1.10%	2.98%

The averages of the deployments, as seen below, show the overall time saved when comparing a single user environment's manual creation to an automated deployment. In total One minute and Fourteen seconds were saved. When compared to a deployment which requires multiple environments for multiple students [such as that of a school environment], you can see how that time saved adds up.

Scenario	***Method***	***Avg Time***	***StdDev Time***	***Avg Peak CPU (%)***	***Avg Peak RAM (%)***	***Time Saved vs Manual***	***Speedup factor***
10 VM	Manual	2:26	34.0s	1.37%	3.28%	-	-
10 VM	Ansible Iteration 1	1:12	0.4s	0.89%	2.99%	1:14	2.02x

Malware Remediation:

WannaCry Artifacts:

Below are all of the artifacts released by the malware sample collected by one Andrei Gorlitsky. The SHA1 hash being “INSERT ID HERE”. This sample was supposedly collected from Malware Bazaar. Proper chain of custody documentation was not created and as such collection information was missed. Future Samples will not have this issue. All artifacts were collected through the use of ProcMon (Process Monitor) a sysinternal tool-suite executable.

Type	*Path / key*	*Detail*	*Significance*
Registry	*HKCU\...\ZoneMap\ProxyBypass*	*REG_DWORD = 1*	*Bypass proxy for C2 reach*
Registry	*HKCU\...\ZoneMap\IntranetName*	*REG_DWORD = 1*	*Trust intranet zone*
Registry	*HKCU\...\ZoneMap\UNCAsIntranet*	*REG_DWORD = 1*	*Lateral movement via UNC*
Registry	*HKCU\...\ZoneMap\AutoDetect*	*REG_DWORD = 0*	*Disables proxy auto-detect*
Registry	*HKCU\Control Panel\Desktop\Wallpaper*	*→ @WanaDecryptor@.bmp*	*Ransom note wallpaper set*

Registry	*HKLM\...\Notifications\Data\418A073AA3BC3475*	*REG_BINARY blob*	*Suppresses security alerts*
File	*...\Wannacry\@WanaDecryptor@.exe*	*Dropped binary*	*Decryptor GUI spawned*
File	*...\Wannacry\@WanaDecryptor@.bmp*	*Ransom note image*	*Set as desktop wallpaper*
File	*...\Wannacry\@Please_Read_Me@.txt*	*Ransom note text*	*Victim-facing demand*
File	*...\Wannacry\Public.KEY*	*RSA public key*	*Used to encrypt AES keys*
File	*Desktop\WannaCry_AES_KEY\r.wnry*	*.wnry key file*	*Encrypted ransomware config*
File	*Desktop\WannaCry_AES_KEY\h.wnry*	*.wnry key file*	*Encrypted ransomware config*
Encrypted	*Wannacry.zip.WNCRY*	*.WNCRY extension*	*Original zip encrypted*
Encrypted	*SysinternalsSuite.zip.WNCRY*	*.WNCRY extension*	*Sysinternals archive encrypted*
Encrypted	*Eula.txt.WNCRY*	*.WNCRY extension*	*Text file encrypted*
Encrypted	*psversion.txt.WNCRY*	*.WNCRY extension*	*Text file encrypted*
Encrypted	*readme.txt.WNCRY*	*.WNCRY extension*	*Text file encrypted*
Encrypted	*Ransomware.wannacry...7z.WNCRY*	*.WNCRY extension*	*Archive encrypted*
Process	*@WanaDecryptor@.exe (PID 6172)*	*Spawned child process*	*Decryptor UI launched*

WannaCry Benchmark:

Scenario	***Method***	***Run***	***Remediation Time***
Scenario 1: WannaCry	Manual	1	27:32
Scenario 1: WannaCry	Manual	2	25:12
Scenario 1: WannaCry	Manual	3	22:23

Scenario 1: WannaCry	Ansible Iteration 1	4	2:57
Scenario 1: WannaCry	Ansible Iteration 1	5	7:56
Scenario 1: WannaCry	Ansible Iteration 1	6	3:35

LokiBot Artifacts:

Type	*Path / key*	*Detail*	*Significance*
File	*C:\Users\windows10\Downloads\Wannacry\Loki\af10c77d2bb3b33acff69640aef7de93.exe*	*MD5 in FileName*	*Primary Payload*
File	*C:\Users\windows10\AppData\Roaming\ABDEE3\38A1E7.exe*	*Hidden, File was copied*	*Persistence Method Located in %appdata%*
File	*C:\Users\windows10\Downloads\Wannacry\Loki\OPENGL32.dll*	*Dropped alongside .exe*	*DLL Hijacking / Evasion*
File	*C:\Users\windows10\Downloads\Wannacry\Loki\CRYPTSP.dll*	*Dropped alongside .exe*	*DLL Hijacking / Evasion*
File	*C:\Users\windows10\Downloads\Wannacry\Loki\WINMM.dll*	*Dropped alongside .exe*	*DLL Hijacking / Evasion*
Registry	*HKLM\HARDWARE\DESCRIPTION\System\CentralProcessor\0\~MHz*	*Read [Continuous]*	*System fingerprinting*
Registry	*HKLM\System\CurrentControlSet\Control\TimeZoneInformation*	*Read [Continuous]*	*Victim geo-profiling*
Registry	*HKLM\SOFTWARE\WOW6432Node\Microsoft\Cryptography\Defaults\Provider Types\Type 001*	*Read*	*Crypto provider lookup for mutex/GUID hash*
Credential Target	*C:\Program Files (x86)\AbleFTP11\...\sshProfiles-j.jsd*	*Probed*	*FTP credential theft*
Credential Target	*C:\Users\windows10\AppData\Roaming\Opera Mail\Opera Mail\wand.dat*	*Probed*	*Email credential theft*
Credential Target	*C:\Users\windows10\AppData\Roaming\Fenrir Inc\Sleipnir\...\Login Data*	*Probed*	*Browser credential theft*
Credential Target	*C:\Users\windows10\AppData\Roaming\Microsoft\Crypto\RSA\S-1-5-21-...\e305bd35...9ad035aa*	*Probed*	*Windows crypto key access*

LokiBot Remediation:

AgentTesla Artifacts:

Type	*Path / key*	*Detail*	*Significance*
File	%AppData%\Roaming\<config-name>\<config-name>.exe	Hidden + System attributes set on copy	Primary persistence method, self-copy to %AppData%
File	%Temp%\<random>.exe	Alternate drop path	Initial staging before persistence established
File	%AppData%\tor.zip	Downloaded from TOR website	TOR proxy for HTTP C2 exfiltration (select variants)
File	<dropped file>:Zone.Identifier (deleted)	ADS removed on first run	Evasion — hides file download origin metadata
File	CO_<username>/<ComputerName><DateTime>.zip	Browser credential archive	Exfiltrated via SMTP as email attachment
File	SC_<username>/<ComputerName><DateTime>.jpeg	Screenshot captured via .NET	Exfiltrated via SMTP as email attachment
Registry	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\<AppName>	Write on first execution	Persistence — launches malware copy on every login
Registry	SYSTEM\CurrentControlSet\Control\Session Manager\AppCompatCache	Read [Shimcache]	Forensic — tracks executables even after deletion
Registry	Amcache.hve	Read [Amcache]	Forensic — links execution to binary hash and path
Process	RegSvcs.exe	Process hollowing target	Hosts injected AgentTesla payload in memory

Process	aspnet_compiler.exe / MSBuild.exe	Alternate hollowing host	Observed in variant campaigns for evasion
Process	SetWindowsHookEx (type 13 — WH_KEYBOARD_LL)	API call on execution	Installs low-level keylogger hook
Process	GetClipboardData	Periodic API call	Clipboard monitoring and exfiltration
Process	Graphics.CopyFromScreen / GDI BitBlt	API call on timer	Desktop screenshot captured as JPEG
Network	Port 25 / 587 outbound SMTP	Continuous exfil channel	Primary C2 method — watch from non-mail processes
Network	Port 20 / 21 outbound FTP (STOR command)	Exfil channel	FTP upload; credentials embedded in binary
Network	HTTP POST to /gate.php or /panel	Exfil channel	HTTP C2; requires attacker web panel
Network	api.telegram.org outbound	Exfil channel	Telegram bot exfiltration — rising in prevalence
Network	SMTP Subject: PW_<user>/<host>	Email subject pattern	Stolen credentials exfil — visible in PCAP
Network	SMTP Subject: KL_<user>/<host>	Email subject pattern	Keylogger data exfil — visible in PCAP
Network	SMTP Subject: SC_<user>/<host>	Email subject pattern	Screenshot exfil — visible in PCAP
Credential Target	%LocalAppData%\Google\Chrome\User Data\Default\Login Data	Probed	Chrome saved passwords (SQLite)
Credential Target	%AppData%\Mozilla\Firefox\Profiles*.default\logins.json	Probed	Firefox credential store
Credential Target	%AppData%\Microsoft\Credentials\	Probed	Windows Credential Manager / Outlook passwords

Credential Target	%AppData%\FileZilla\recentservers.xml	Probed	FTP credential theft — recent server connections
Credential Target	%ProgramFiles%\OpenVPN\config\	Probed	VPN credential theft
Credential Target	%AppData%\NordVPN\	Probed	VPN credential theft
Credential Target	%AppData%\Opera Mail\Opera Mail\wand.dat	Probed	Email credential theft
Credential Target	Clipboard (live)	Probed [Continuous]	Captures copied passwords, tokens, card numbers

AgentTesla Remediation:

Kovter Artifacts:

Kovter Remediation:

Citations:

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